Jake Joyner

Software Engineer | Backend, Infrastructure and Security | LLM Evaluation and AI Code Review

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SUMMARY

Co-founder and founding engineer of AILedger, PBC, a deployed open-source audit-logging platform for AI-influenced decisions; shipping production TypeScript and Python since 2024. Daily work is specifying, reviewing, testing and grading code written by LLM coding agents (3,900+ commits reviewed across 80+ repositories in 2026, 149 merged pull requests authored) and building the evaluation harnesses, hidden-oracle benchmarks, pairwise-preference tools and LLM-judge gates that measure them. Owns and operates a self-hosted identity, secrets and transparency-log stack. Open to remote contract and part-time AI training, evaluation and data-quality work.

EXPERIENCE

Co-Founder and Founding Engineer, AILedger, PBC, Los Angeles, CA | Mar 2026 - Present

Co-founded with Stanley Bishop (UCLA). Audit substrate for AI-influenced decisions, built for Federal Rule of Evidence 707 and EU AI Act Articles 12, 19 and 26: proxy.ailedger.dev, dash.ailedger.dev, github.com/ailedger-dev/ailedger (AGPLv3; detection layer Apache-2.0).

- Designed and built the full codebase (about 41,000 lines of TypeScript and Python, 286 commits): Cloudflare Workers edge proxy (Hono), hash-chained PostgreSQL ledger on Supabase with integrity enforced by database triggers across 20 migrations, TypeScript SDK, Python CLI, React dashboard; about 360 tests across 5 suites (vitest, pytest, fast-check); CI enforces byte-exact RFC 8785 canonicalization parity between languages; GitHub Actions with PR-preview, main and tag promotion; HMAC pseudonymization with per-tenant salts.
- Reviewed and merged 3,900+ commits across 80+ repositories in 2026, most authored by LLM coding agents working from my specifications: verified behavior against spec, ran and extended tests, rejected or rewrote for correctness, efficiency and clarity, and documented the reasoning. Authored the engineering guidelines the agents work from.
- Designed and run the company's self-hosted platform (about 15 services under rootless podman): Keycloak passkey single sign-on (OIDC), Forgejo plus CI, OpenBao secrets with an SSH certificate authority issuing one-hour certificates, Nebula and NetBird mesh, Prometheus and Grafana; origin locked to Cloudflare with Authenticated Origin Pulls after a study of Anthropic, OpenAI, xAI and Mistral's public identity surfaces; off-site backups with passed restore drills.
- Put a tamper-evident operational record into production: per-tenant hash chains, Ed25519 node signatures, RFC 6962 signed tree heads, daily checkpoints mirrored to an independent witness repository, and a standard-library-only public verifier (verify.ailedger.dev); its threat model became 22 automated daily integrity checks (1,078 passing runs, failures open alarm cards).
- First-authored 9 of the company's 15 governance ADRs and merged 20 pull requests from my co-founder's agent desk; ran formation and compliance pro se (Delaware PBC, California qualification, public Charter v1.5, stock plan, provisional patent and trademark filings) and automated the compliance calendar against a 23-item filing register.

Maintainer and Org Admin, aDNA Network (90 repositories), Remote | Jul 2026 - Present

Open standard and reference vaults for aDNA (Agentic DNA), a knowledge-graph architecture shared by people and AI agents: adna.network, github.com/aDNA-Network.

- Designed and shipped the cross-platform node installer and enrollment system in 17 days: stdlib-only Python core, PowerShell installer, C# Windows launcher, macOS, AUR, Homebrew and a self-hosted pacman repo; 25 pinned releases across 6 channels, live today. Signed-release supply chain (minisign, pure-stdlib RFC 8032 verifier, SHA-256 pins); a Cloudflare Worker enrollment queue on a SQLite Durable Object that holds no signing keys, with a signed single-use invite format; 16-check conformance suite; tested on a 4-distro container matrix, QEMU VMs, Windows 11, macOS and a Raspberry Pi, catching a production-breaking Cloudflare 403 that CI missed; 3 GitHub Actions workflows.
- Wrote the network's trust-tier threat model (ADR-022, four certificate-group tiers closing a client-to-CA pivot); deployed OpenBao fleet secrets with a 3-share threshold-2 unseal; onboarded partner, Windows, Ubuntu and GPU nodes onto a 19-node mesh. Live aDNA demo at a Claude Code user meetup, Pasadena (Aug 2026); panelist, The KINN "AI Cafe" (Jul 2026).

LLM Evaluation and Agent Tooling (independent, open source where noted) | 2024 - Present

- Built the experiment harness for a context-engineering study: paired headless Claude Opus arms build a 53-requirement site in isolated git worktrees at a pinned commit, graded by a hidden 53-check oracle with holdout data and per-turn token metering; published every voided run and 22 documented defects. Result: arms indistinguishable on cost (0.1% apart against an 8.8% run-to-run spread).
- Built an overnight adversarial-verification harness for coding agents: hypotheses pre-registered with hash-sealed success criteria, every finding reproduced by the supervisor in a fresh worktree and challenged by three independent refuter agents under call, token and dollar caps; first night 21 tasks, 18 findings stood (1 critical, 3 high), 3 refuted, $65.

- LLM-judge quality gates (independent model, fresh context, 164 graded runs); a 22-rule behavioral spec for an email-drafting agent scored 24/26 by a fresh agent with refusal and send-trap tests passed; pairwise-preference tool with Bradley-Terry ranking over 120 judgments. Independent RFC 6962 transparency-log monitor (33 tests, mypy --strict) that caught a tampered production ledger; codex-vs-claude-bench; PromptPotential rubric scorer (2024).
- Production pipelines on systemd across two hosts: voice-to-action (13.9k lines stdlib Python; typed action extraction, human approval gate, kill switch, circuit breaker, 2,089 hash-chained ledger events); comms and video ingestion (BlueBubbles, mail, WhatsApp via a 712-line Go client, yt-dlp, whisper, gitleaks gate; 199 videos); a YouTube demand scanner with 326 tests; a Twilio voice agent on a hand-rolled RFC 6455 WebSocket server.
- Systems: Endurance, multi-agent dev environment for coding agents (77-verb CLI, 137 skills, 1,253 commits, 2026); operations-bridge, FastAPI claim-lease daemon with fencing tokens, co-developed (602 tests); skillrt Rust runtime on crates.io; sys-update, agent-reviewed Arch update pipeline; home-server hardening (four-zone nftables, systemd exposure 9.4 to 1.6, rootless podman, tested restores); 190 systemd units across laptop and server; 321-line Rust patch to lan-mouse.

Hackathon, "Agents, Everywhere" Global Hackathon (AI Tinkerers LA / OpenAI), The KINN | Sep 2026

Top contributor (71 of 103 commits in about 5 hours) on a four-person team building a smart-glasses social copilot: owned the persistent person-memory service (TypeScript, node:sqlite) and corroborated research write-back; submitted and demoed same day (github.com/reimagen/mmg).

Earlier: *Undergraduate Researcher, T-REx program, UC San Diego Mathematics (Aug 2025): optimization methods in machine learning with Prof. Lijun Ding, research poster. Computer Engineer, NASA Community College Aerospace Scholars (Dec 2024 - Apr 2025): computer systems on a 10-person Mars-mission-simulation team; invited to Mission 3. Founder, Fly Abstract (Apr 2023 - Dec 2024): FAA Part 107 drone video business with paying clients. Restoration Technician, True West Motorworks (Sep 2025 - Apr 2026).*

SKILLS

Languages: Python, TypeScript, JavaScript, SQL, Bash, Rust, Go, PowerShell; C and C++ (coursework), C#

AI / LLM: LLM evaluation, benchmark and hidden-oracle design, rubric design, LLM-as-judge gates, RLHF-style pairwise comparison, Bradley-Terry ranking, response grading, data annotation and quality review, prompt engineering, agent behavioral specs, agentic coding tools (Claude Code, Codex), MCP server development, token metering, llama.cpp local inference, whisper

Security and verification: hash chains, Merkle trees, RFC 6962 transparency logs, Ed25519, minisign, RFC 8785 canonical JSON, RFC 3161 timestamps, HMAC pseudonymization, threat modeling, Keycloak / OIDC / passkeys, OpenBao, SSH certificate authorities, Cloudflare Authenticated Origin Pulls, nftables, systemd sandboxing, gitleaks

Backend, data and testing: REST APIs, Hono, FastAPI, Pydantic, WebSockets, PostgreSQL, SQLite, Supabase, migrations and triggers, JSON Schema, React; code review, debugging, unit, integration, property-based (fast-check), golden-vector and adversarial testing, vitest, pytest, mypy --strict, ruff, GitHub Actions, Forgejo Actions

Infrastructure: Cloudflare Workers, Pages, KV, R2, Durable Objects, Tunnel; Linux (Arch, Ubuntu, systemd), Docker, rootless podman, Nebula, NetBird, Tailscale, Caddy, Prometheus, Grafana, Forgejo, Git, Dolt, Hedera SDK, Twilio, QEMU

EDUCATION

University of California, San Diego, B.S. Mathematics-Computer Science (in progress) | Sep 2025 - Jun 2027 (expected)

James R. Bunch Mathematics Endowed Scholar (2026-27). Provost Honors (Fall 2025). IEEE-Eta Kappa Nu. Coursework: abstract algebra, combinatorics, extremal graph theory, optimization, systems programming in C.

Santa Monica College, Computer Science, 3.9 GPA, Scholars Program, Dean's Honor List | Feb 2023 - Jun 2025

PUBLICATIONS, PATENTS AND CERTIFICATIONS

Co-author with Stanley Bishop and Luke Waltman, "From Retrieval to Convergence: Evolving Context Systems for LLM Reasoning", preprint, Sep 2026. Sole inventor, U.S. provisional patent application 64/090,070 (decision-memorialization for multi-agent systems), Jun 2026. Panelist, "Who Owns Your AI Agent?", The KINN AI Cafe, Jul 2026 (youtube.com/watch?v=BI6rv6HzUuI). Essays on AI governance at jakejoyner.com (May-Jun 2026). FAA Part 107 Remote Pilot Certificate, Jun 2023.